

Intermediate Macro

Chapter 15: Expectations, Consumption, and Investment

Instructor: Mihyun You

Department of Economics
Georgetown University

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GEORGETOWN UNIVERSITY

- 15.1 Consumption
- 15.2 Investment
- 15.3 The Volatility of Consumption and Investment



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- Chapter 15 introduces the terms **permanent income theory of consumption** and **life cycle theory of consumption**.
- **Human wealth** is the present value of expected after-tax labor income.
- **Tobin's q** is the ratio of a firm's financial value—the value of existing stock plus the value of bonds outstanding—to the replacement cost of the firm's capital. Theory and evidence suggest that Tobin's q should be positively related to investment.
- The **user** or **rental cost of capital** is the sum of the real interest rate and the depreciation rate on a unit of capital.

Modern Theory of Consumption

15.1 Consumption

- In earlier chapters, consumption is described as a function solely of current disposable income.
- In fact, however, people plan over longer horizons and are willing to borrow to finance current consumption when current disposable income is temporarily low.
- The modern theory of consumption was developed independently by two prominent economists.
- Milton Friedman called the theory he developed the **permanent income theory of consumption** which emphasized that consumers look beyond current income.
- Franc Modigliani called his theory the **life-cycle theory of consumption** which emphasized consumers' natural planning horizon in their entire lifetime.

The Very Foresighted Consumer

15.1 Consumption

- To understand both theories, assume that a person wants a constant flow of consumption over her lifetime.
- In this case, a perfectly rational person would develop a consumption plan in two steps.
 - ① First, she would calculate her **total wealth**—assets on hand (**financial** and **housing wealth**) plus the present value of future labor income (so-called **human wealth**).
 - ② Then, she would calculate the proportion of this wealth that should be spent each year to maintain a constant consumption level over her lifetime. If it happened that this level of consumption fell short of current income, the difference would be borrowed.
- What we have described is a consumption decision of the form

$$C_t = C(\text{total wealth}_t) \quad (15.1)$$

Toward a More Realistic Description

15.1 Consumption

- In practice, most consumers following such a plan would end up borrowing large sums of money early in life, because income during college and early working years is likely to be very low relative to income later in life.
- In fact, however, most young adults do not borrow the relatively large sums suggested by simple calculations, for several reasons.
 - ① First, they may not intend to maintain constant consumption over their lifetimes.
 - ② Second, the computations involved in planning for constant consumption may be too complicated.
 - ③ Third, human wealth is based on forecasts of future earnings, which may turn out to be less than expected.
 - ④ Fourth, banks may be unwilling to extend much credit to young adults on the expectation of future earnings.

Consumption Function Depending on Two Factors

15.1 Consumption

- This discussion suggests that consumption is likely to depend on two factors:
 - ① wealth—because consumers are to some degree forward looking—and
 - ② current disposable income—because consumers may be unwilling to or unable to calculate and implement a spending plan expected to maintain constant consumption over their lifetimes.
- We can build a consumption function of the form:

$$C_t = C \left(\begin{array}{c} \text{total wealth}_t, \\ (\quad + \quad , \quad + \quad) \end{array} Y_{Lt} - T_t \right) \quad (15.2)$$

where Y_{Lt} denotes real labor income in year t .

Do People Save Enough for Retirement?

15.1 Consumption

- Evidence on retirement saving suggests that most consumers save sufficient resources for retirement.
- This finding lends support to the importance of wealth (and therefore expectations) in consumption behavior.
- On the other hand, a substantial fraction of households (about 20% in some studies) do not save enough for retirement.
- For many of these households, the present value of Social Security benefits accounts for almost all of their retirement wealth.

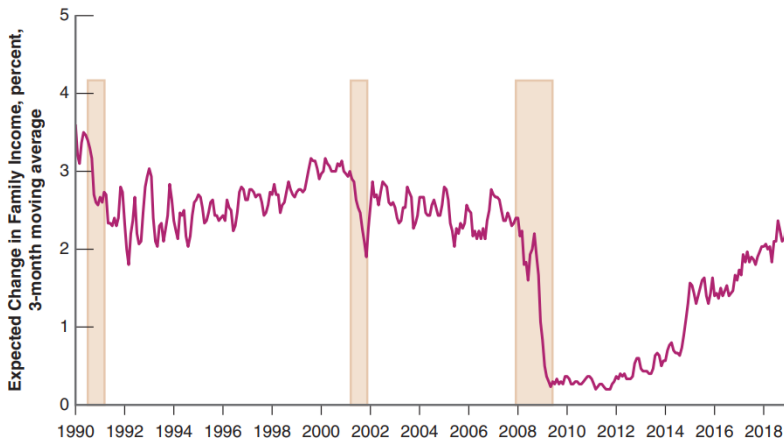
Putting Things Together: Current Income, Expectations, and Consumption

15.1 Consumption

- The fact that consumption depends upon wealth (which in turn depends upon expectations about the future) has two empirical implications.
 - ① First, fluctuations in current income are likely to generate less than proportional fluctuations in consumption.
 - ② Second, consumption can be affected by changing expectations about the future, even when current income does not change.
- The effects of the Great Recession are particularly striking in this respect.

Expected Change in Family Income since 1990

15.1 Consumption



Investment and Expectations of Profit

15.2 Investment

- When deciding whether to purchase a new machine or to build a new plant, firms compare the expected present value of profit from the machine or plan to the cost.
- If the present value of profit exceeds the cost, they invest; if not, they do not invest.
- The calculation of the expected present value of profits requires not only a forecast of profit, but also a consideration of the wear and tear on the machine or plant from use.
- Wear and tear is called depreciation.

The Present Value of Expected Profits

15.2 Investment

- To capture the fact that it takes some time to put machines in place (and even more time to build a factory or an office building), let us assume that a machine bought in year t becomes operational—and starts depreciating—only one year later, in year $t + 1$.
- Denote profit per machine in real terms by Π .
- If the firm buys a machine in year t , the machine will generate its first profit in year $t + 1$; denote this expected profit by Π_{t+1}^e .
- The present value, in year t , of this expected profit in $t + 1$ is given by

$$\frac{1}{1 + r_t} \Pi_{t+1}^e$$

The Present Value of Expected Profits

15.2 Investment

- Denote expected profit per machine in year $t + 2$ by Π_{t+2}^e .
- Note that because of depreciation, only $(1 - \delta)$ of the machine is left in year $t + 2$, so the expected profit from the machine is equal to $(1 - \delta)\Pi_{t+2}^e$.
- The present value of this expected profit as of year t is equal to

$$\frac{1}{(1 + r_t)(1 + r_{t+1}^e)}(1 - \delta)\Pi_{t+2}^e$$

- The same reasoning applies to expected profits in the following years.
- The present value of expected profits from buying the machine in year t , which we shall call $V(\Pi_t^e)$:

$$V(\Pi_t^e) = \frac{1}{1 + r_t}\Pi_{t+1}^e + \frac{1}{(1 + r_t)(1 + r_{t+1}^e)}(1 - \delta)\Pi_{t+2}^e + \dots$$

The Investment Decision

15.2 Investment

- To simplify notation, let us assume the real price of a machine equals 1.
- What the firm must do is to compare the present value of profits to 1.
- Now let us go from this one-firm, one-machine example to investment in the economy as a whole.
- Denote profit per unit of capital for the economy as a whole by Π_t .
- Our discussion suggests an investment function of the form

$$I_t = I[V(\Pi_t^e)] \quad (15.4)$$

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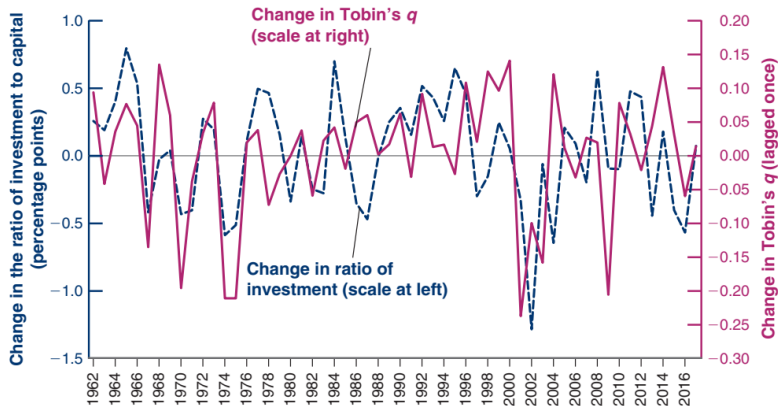
Investment and the Stock Market

15.2 Investment

- James Tobin pointed out that firms could use information available in financial markets to simplify the investment decision.
- The financial value of a firm (its stock market plus the value of bonds outstanding) measures the value financial investors place on capital (plant and equipment) already in place.
- Firms should invest when the financial value of a unit of their capital exceeds the cost of an additional unit of capital.
- If firms behave in this way, there should be a positive relationship between aggregate investment and the ratio of the total financial value of firms to the replacement cost of their capital.
- The latter ratio is called **Tobin's q** .

Tobin's q versus the Ratio of Investment to Capital

15.2 Investment



A Convenient Special Case

15.2 Investment

- It is useful to go through a special case where the relation between investment, profit, and interest rates becomes simple.
- Suppose firms expect both future profits (per unit of capital) and future interest rates to remain at the same level as today, so that

$$\Pi_{t+1}^e = \Pi_{t+2}^e = \cdots = \Pi_t$$

and

$$r_{t+1}^e = r_{t+2}^e = \cdots = r_t$$

- Economists call such expectations—that the future will be like the present—**static expectations**.

User cost or Rental Cost of Capital

15.2 Investment

- Under these two assumptions, equation (15.3) becomes

$$V(\Pi_t^e) = \frac{\Pi_t}{r_t + \delta} \quad (15.5)$$

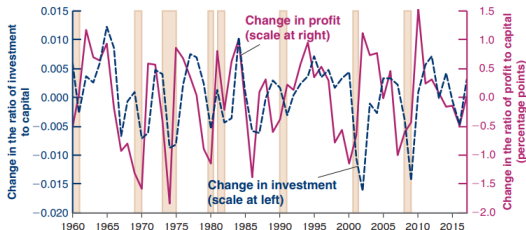
- Replacing (15.5) in equation (15.4), investment is given by:

$$I_t = I \left(\frac{\Pi_t}{r_t + \delta} \right) \quad (15.6)$$

- The sum of the real interest rate and the depreciation rate is called the **user cost** or the **rental cost** of capital.
- Even though firms typically do not rent the machines they use, $(r_t + \delta)$ still captures the firms' implicit cost—sometimes called the *shadow cost*—of using the machine for one year.

15.2 Investment

- Theory implies that investment should depend on expected future profit, but there is also evidence that investment moves strongly with fluctuations in *current profit*.



- In summary, the investment equation is better specified as

$$I_t = I[V(\Pi_t^e), \Pi_t] \quad (15.7)$$

Profit and Sales

15.2 Investment

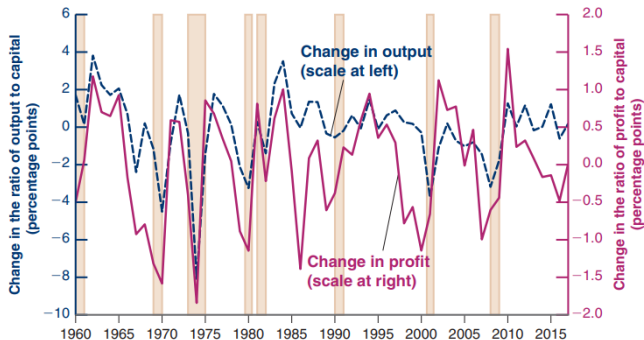
- Investment depends on current and expected future profit, but what determines profit? Primarily two
 - the level of sales and
 - the existing capital stock.
- The level of profit per unit of capital is likely to be closely related to the level of sales per unit of capital.
- Ignoring the distinction between sales and output, sales per unit of capital can be proxied by output per unit of capital.
- Let Y_t denote output (equivalently, sales), and K_t the capital stock at time t .
- Our discussion suggests the following relation:

$$\Pi_t = \Pi\left(\frac{Y_t}{K_t}\right)_{(+)} \quad (15.8)$$

Changes in Profit per Unit of Capital versus Changes in the Ratio of Output per Capital in the United States since 1960

15.2 Investment

- In fact, there is a close relationship between changes in profit per unit of capital and changes in the output-capital ratio.



Similarities and Differences between Consumption and Investment

15.3 The Volatility of Consumption and Investment

- Although the consumption and investment decisions have some similarities, the theory developed above suggests that investment is much more volatile than consumption.
- After an increase in income perceived as permanent, consumers respond with at most an equal increase in consumption.
- After an increase in sales perceived as permanent, however, firms may respond by investing in projects many times larger than the increase in sales.
- In the absence of adjustment costs, firms have no reason to maintain a smooth flow of investment.
- Once projects become profitable, firms invest immediately.
- Consumers, on the other hand, desire to maintain a relatively constant level of consumption.

Rates of Change in Consumption and Investment in the United States since 1960

15.3 The Volatility of Consumption and Investment

- The movements of investment are much higher than those of consumption in percentage terms.

